



ADITI BHAT

Biomolecular & Biomedical Engineering | R&D | Biotech & Medtech

San Francisco, CA | 415-629-7980 | aditibhat017@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

B.S. Biomolecular Engineering & Bioinformatics

University of California, Santa Cruz

Sep 2022 – Dec 2025

4.0 Major GPA | Honors at Graduation |

Baskin Engineering Dean's Award (2026)

Sensor Technologies for Biomedical Applications Specialization

Indian Institute of Science, Coursera

Aug 2026 - Sep 2026

TECHNICAL SKILLS

- **Molecular Biology:** PCR, Gibson Assembly, Golden Gate Assembly, primer/plasmid design, bacterial culture, gel electrophoresis, fluorescence-based assays, aptamer characterization
- **Protein Engineering:** Protein expression, purification, assay design, RFdiffusion, AlphaFold3, PyMOL
- **Computational & Data Analysis:** Python, R, MATLAB, statistical modeling, regression, cross-validation, feature engineering, data visualization
- **Bioengineering:** Wearable biosensors, sensor design, biomechanical analysis, anatomical mapping, 3D printing, Fusion 360/CAD
- **Tools:** Benchling, Git/GitHub, Google Colab

SUMMARY

Biomolecular Engineering & Bioinformatics graduate combining hands-on biological research, quantitative analysis, and bioengineering design to translate health challenges into testable solutions. Brings experience in molecular biology, biosensors, computational analysis, and interdisciplinary project leadership from early-stage development through scientific presentation.

RESEARCH EXPERIENCE

Undergraduate Research Team Lead - Bernick iGEM Lab

University of California, Santa Cruz | Aug 2024 – Nov 2025

- Led an 18-member interdisciplinary team across 4 wet-lab and 1 dry-lab subteams developing **SafeTEA**, an aptamer-based platform designed for aflatoxin B1 capture.
- Developed recombinant constructs using primer/plasmid design, Gibson Assembly, and Golden Gate cloning and iteratively optimized fluorescence-based aptamer assays for candidate characterization.
- Created proof-of-concept evidence supporting candidate target interaction and dual-aptamer system development; presented work at the 2025 iGEM Grand Jamboree in Paris, earning Silver Medal.

RELEVANT PROJECTS

"GaitMate" - Wearable Biosensor for Ligament Rehabilitation

Faculty-Sponsored Bioengineering Project | Sep 2025 – Present

- Led 4–6 member team translating unmet rehabilitation need into an early-stage wearable biosensing platform for lateral ankle ligament rehabilitation across ATFL, CFL, and PTFL.
- Designed 9-location anatomical sensing framework and strain-analysis approach incorporating maximum strain, spatial gradient, and signal variability for longitudinal rehabilitation monitoring.
- Conducted clinical/customer discovery and early commercialization; earned 2nd place among ~40 teams in iGEM Venture Creation Lab; selected as 1 of 4 startup pitches at 2026 UC Bioengineering Symposium.

PUBLICATIONS

- Bhat, A.A., & Hendricks, D.G. "Student Leadership Experiences in Competitive Engineering Research Teams." ASEE Zone IV Conference, 2026. doi:10.18260/1-2--58431
- Bhat, A.A., Rocha, L.R., et al. "Designing a Wearable Biosensor for Ligament Rehabilitation." ASEE Zone IV Conference, 2026. doi:10.18260/1-2--58406

SELECTED SCIENTIFIC PRESENTATIONS

- UC Bioengineering Symposium, 2026: Wearable Biosensing Poster & Startup Pitch (1 of 4 selected)
- ASEE Zone IV Conference, 2026: 2 Oral Presentations + Poster
- iGEM Grand Jamboree, Paris, 2025: Live Talk, Poster & Technical Wiki

AWARDS & HONORS

- Baskin Engineering Dean's Award (2026)
- 2nd Place, iGEM Venture Creation Lab Biofoundation Program (2026)
- Cynthia Mathews Research Scholarship (2025)
- iGEM Silver Medal (2025)
- Cum Laude at Graduation (2025)

LANGUAGES

- English (Native)
- Japanese (Conversational)
- Italian (Conversational)
- Hindi (Elementary)
- Chinese (Elementary)

RELEVANT PROJECTS CONTINUED

Digital Biomarkers for Cognitive Decline

Independent Computational Bioengineering Project | 2026

- Integrated wearable, cognitive, and demographic NHANES data into a harmonized 1,289-participant cohort and developed Python pipelines for feature engineering, regression, 5-fold cross-validation, and cross-year generalization.
- Identified an association between mean activity and cognitive performance ($r = 0.174$, $p < 10^{-10}$), with wearable activity improving cross-validated model performance from $R^2 = 0.379$ to 0.386 .

Endometriosis Digital Health & Symptom Tracker

Independent Computational Bioengineering Project | 2026

- Developed a longitudinal digital-health framework integrating menstrual symptoms, pain, sleep, activity, HR/HRV, and wearable-derived physiological measures to investigate personalized patterns associated with endometriosis symptoms.

TEACHING & LEADERSHIP EXPERIENCE

Head Teaching Assistant - BME 5, 21L, 22L, 23L, 185

Baskin Engineering, UC Santa Cruz | Jan 2024 – Dec 2025

- Supported biotechnology, wet-lab, and technical-writing instruction in PCR, gel electrophoresis, bacterial culture, experimental troubleshooting, data interpretation, and scientific communication.
- Mentored students through laboratory work, technical reports, theses, and presentations while supporting laboratory safety and EHS compliance.

Undergraduate Engineering Peer Advisor

Baskin Engineering, UC Santa Cruz | Aug 2023 – Dec 2025

- Advised students on degree planning and academic policies and contributed student/faculty feedback toward redesign of Biomolecular Engineering degree pathways.